

What happens when women suddenly stop receiving cash transfers?*

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Abstract

Female-targeted cash transfers are widely used as a policy tool to enhance women's empowerment. However, little is known about what happens when payments stop – whether due to budget cuts, program changes, or recipient graduation. We study how women are affected by unanticipated program exit, and whether they experience backlash when they stop bringing cash home. Using a regression discontinuity-in-differences design around a revised eligibility threshold, we follow 2,333 women exiting Pakistan's largest cash transfer program, surveying them before program exit and one year after it. Drawing on comprehensive measures of empowerment, intimate partner violence, and well-being, we find no evidence of negative impacts a year after exit. These results suggest that the gains experienced by women during the program are not undone by adverse reactions upon exit.

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1 Introduction

Large-scale cash transfer programs enjoy tremendous popularity in low and middle-income countries, reaching more than 1.9 billion people globally ([World Bank, 2015](#)). Beyond the goal of alleviating poverty, many programs are deliberately targeted toward women, seeking to increase their control over resources, strengthen their household bargaining power, and improve their well-being. Indeed, in contexts with low female labor force participation, a female-targeted cash transfer may dramatically raise her share of household income, even when the absolute amount is modest. A large body of work has examined the impact of female-targeted transfers, with many studies finding evidence of improvements in women’s decision-making power, empowerment, and reduced gender-based violence (see [Bastagli *et al.* \(2019\)](#); [Buller *et al.* \(2018\)](#); [Banerjee *et al.* \(2017\)](#); [Diaz-Pardo and Rao \(2024\)](#); [Crosta *et al.* \(2024\)](#) for reviews). Evidence also suggests that these transfers shift household spending patterns, increasing the share of expenditures on food and children’s goods ([Angelucci and Attanasio, 2013](#); [Hidrobo *et al.*, 2016](#); [Armand *et al.*, 2020](#)).

However, a recipient may not receive cash transfer payments indefinitely. She may exit a transfer program due to changes in program design, funding reductions, or improvements in household economic conditions leading to program “graduation”.¹ The effects of exiting a cash transfer program on women’s empowerment are ex ante ambiguous. Once transfers stop, a major concern is the potential for backlash if other family members blame the recipient for the discontinuation of funds or suspect her of diverting the money received. This might be especially likely when program exit only occurs among a subset of beneficiaries, while others in the community continue to receive the transfers – raising suspicion that the woman is concealing money, and potentially triggering extractive violence ([Haushofer *et al.*, 2019](#); [Baranov *et al.*, 2021](#)).² In addition, if cash transfers improve women’s status merely through control over additional income, then the benefits could dissipate once a household exits the program – leading to a deterioration in female empowerment back to pre-transfer levels. Conversely, repeated cycles of receiving, withdrawing, and spending the transfer may serve as a learning opportunity for the household: as the woman and her relatives observe her consistent control over the funds, they gradually update their beliefs about her financial agency and literacy, leading to

¹[Medellín *et al.* \(2015\)](#) report in a review of Latin American transfer programs that almost all countries had initiated a process of “recertification” to update the recipient lists, similar to the program we study in Pakistan; more recently, in a meta analysis of 53 recurring unconditional cash transfers reviewed in a recent global meta-analysis by [Crosta *et al.* \(2024\)](#), almost a third were classified as “completed stream,” i.e. payments had ended.

²Some studies find that cash transfers lead to an increase in violence for certain sub-groups, potentially due to extractive violence ([Angelucci, 2008](#); [Hidrobo and Fernald, 2013](#); [Heath *et al.*, 2020](#); [Mahmud *et al.*, 2025](#)). In our setting, we do not study the receipt of the cash transfer, but rather program exit for some but not others.

sustained empowerment over time. Therefore, generating evidence on how female recipients are affected after cash transfers stop is crucial.

In this paper, we estimate the impact on female recipients of exiting one of the world's largest unconditional cash transfer programs in Pakistan, the Benazir Income Support Programme (BISP). This cash transfer was delivered to the same group of recipients for over a decade, with documented positive impacts on women's empowerment outcomes, including physical mobility, autonomy in decision-making, personal savings, political participation, and a reduction in some forms of gender based violence (Cheema *et al.*, 2016, 2020; Ambler and de Brauw, 2024). In 2022, BISP revised its eligibility cutoff based on a new census conducted to update the data in the Proxy Means Test used for targeting the transfer. These changes led to some previously eligible recipients becoming ineligible and exiting the program. In our study district, Layyah, the reform led 24% of recipients to become ineligible.³ Importantly, other beneficiaries in the same communities who were below the updated cutoff continued to receive transfers. In this context, we use a regression discontinuity-in-difference approach to examine the impacts of program exit on recipients just above the new eligibility cutoff, who were thus discontinued, compared to recipients just below the cutoff, who continued to receive the transfers. We drew our sample using a 3.5-point bandwidth around the revised eligibility cutoff defined by a Proxy Means Test. We verify that this resulted in the two groups of recipients being similar at baseline on observable characteristics.⁴ Within this bandwidth, we sampled 2,449 households from Layyah district in southern Punjab, Pakistan.⁵ Access to administrative Proxy Means Test data allowed us to draw a sample precisely selected to enable our identification strategy: those who were receiving transfers at baseline and who fell just above or just below the new cutoff. We coordinated with BISP and conducted a baseline in-person survey with these households before the newly exited beneficiaries stopped receiving transfers and - crucially - before they knew that they would stop receiving the transfers. The majority (78%) of the respondents at baseline stated they expect to continue receiving payments in the future.

Because our sample consists of women in households close to the cutoff for program exit, by definition, they are not the poorest of the poor in Pakistan. Thus, the transfer represents a relatively modest share of

³This did not represent a cut back in the program budget; many new recipients enter the program during this change, but these recipients are not the focus of our study. Moreover, the formula stayed similar over time apart from the addition of geographic variables to more accurately predict consumption (Guven *et al.*, 2024). Thus, the exit of households in our "treatment" group is a result of both some changes in households' poverty levels over time, variation due to the adjustments to the formula, as well as variations in measurement between rounds. We show in Section 5.2 that results are not sensitive to whether a recipient's PMT score had increased between the two rounds of Proxy Means Tests.

⁴The new Proxy Means Test score ranges from 0.32 to 83.4 for the sample district. Approximately 46,000 households in Layyah were below the old cutoff. Of these, nearly 28% fall within this 3.5 bandwidth around the new cutoff.

⁵Budgetary constraints determined our sample size, and the corresponding bandwidth.

expenditures: approximately 15% at baseline. However, 88% of the women in our sample are not employed at baseline; thus the BISP transfer likely represents the entirety of income brought into the household by the vast majority of women in our sample.

We followed up with our sample with a survey in early 2023, about one year after the exit of newly ineligible beneficiaries. By this point, we expect that households would have had sufficient time to become aware of their program exit and for beneficiaries to register the loss of income, making potential impacts more likely to be detectable. We successfully re-surveyed 95% of the baseline respondents, resulting in a balanced panel of 2,333 observations.

By the time of follow-up, administrative data confirm that newly ineligible households had missed two payments of PKR 14,000 (US\$ 303 PPP).⁶ Additionally, 87% of these respondents were aware that they were no longer beneficiaries, but 51% said they did not know *why* they had been exited from the program — suggesting limited official communication from BISP. This environment creates a high potential for backlash effects, as a husband might blame his wife for her exit from the program, or even think that his wife is in fact continuing to receive the transfer but hiding it, leading to extractive violence.⁷

We find that program exit had no detectable impact on a broad set of indicators of women's empowerment and well-being. These include economic participation on the extensive and intensive margin; measures of physical mobility; a comprehensive set of measures of intimate partner violence, including Audio-Computer Assisted Self-Interview (ACASI) measures (Cullen, 2023); an extensive set of measures of decision-making power, following Jayachandran *et al.* (2023); stated willingness to pay to hide income from the respondent's spouse; a series of subjective well-being measures; and a standardized mental health scale (PHQ-9). Based on the confidence intervals of our estimates, we can rule out impacts larger than 0.12 SD on key indices of women's empowerment. Contrasted with the findings of previous well-identified studies finding positive impacts of the BISP cash transfer on a range of female empowerment outcomes (Cheema *et al.*, 2016, 2020; Ambler and de Brauw, 2024), these null results are striking. The results are not sensitive to specification or sample choices and also hold across a large number of subgroups for whom a stronger negative effect or backlash due to program exit might be expected. We also see no change in the allocation of expenditure

⁶At this time, the payments were being made biannually. The bi-annual payments were each PKR 14,000 – in effect from March 2022 (FY 2021/22) – totalling PKR 28,000 for a year. The bi-annual transfer amount was subsequently increased to PKR 17,500 effective January 2023, and remained at this level during our follow-up survey.

⁷In our setting, the former channel for backlash is more likely, as enumerator accounts confirm that husbands have the same information about payment discontinuation as their wives, because women in our sample area almost exclusively travelled with their husbands to collect the payment.

towards women and children, including educational expenditure and enrollment and assignable expenditures on women's and children's clothing and high-quality food consumption.

Our survey was not designed to capture impacts on overall household economic outcomes, which we expect to be more subtle than the gendered impacts of program exit. This is because while the transfer represents the entirety of female income in most cases, it accounts for only 15% of household expenditures and 7% of consumption. We measure household expenditure using a single survey question rather than a detailed consumption module, yielding a crude and noisy proxy. We find no significant impact of program exit on this outcome. This null result likely reflects a combination of two factors: the transfer's modest share of total household expenditure limits the scope for detectable effects, while measurement error may attenuate our estimates further. It is also possible that other household income sources expanded in parallel with — or in response to — the loss of BISP payments. Without more granular income and consumption data, we cannot disentangle these mechanisms.

To our knowledge, this is the first study to track a panel of female cash transfer recipients where a subset are removed from the program unexpectedly. This design enables us to precisely identify the effects of program exit on women's empowerment and well-being. Such a case has not previously been examined empirically, likely because of the challenges involved in identifying the target sample and timing data collection round an exit event. The existing literature examines the impacts of cash transfers on female empowerment, comparing recipients to non-recipients in the short run (see [Bastagli et al. \(2019\)](#); [Buller et al. \(2018\)](#); [Banerjee et al. \(2017\)](#); [Diaz-Pardo and Rao \(2024\)](#) for reviews). A smaller fraction of this work examines the persistence of the effects of cash transfers ([Kondylis and Loeser, 2021](#); [Crosta et al., 2024](#); [Grisolia, 2024](#)), — and in particular the effects on past recipients of unconditional regular transfers, which [Crosta et al. \(2024\)](#) refer to as the effects of a “completed stream.” These studies have limited coverage of gendered outcomes. For example, a review by [Grisolia \(2024\)](#) finds that only two studies examine the persistence of unconditional cash transfer impacts on IPV.^{8,9}

Importantly, all these studies identify effects by comparing past recipients of transfers to never-recipients. Such comparisons between past recipients and never-recipients necessarily bundle any accumulated benefits

⁸[Haushofer and Shapiro \(2018\)](#) document an insignificant decrease in IPV three years after lump-sum cash transfers in Kenya, while [Roy et al. \(2019\)](#) document a reduction in IPV for women who receive both transfers and behavior change communication 6 months after transfers end in Bangladesh.

⁹A number of papers covered in [Grisolia \(2024\)](#) examine the effects of transfers targeting adolescents, such as conditional cash transfers for education, on recipients' outcomes later in life, quite distinct from the effect of program exit from unconditional cash transfers.

from the program with potential backlash or other unintended consequences due to program exit. In contrast, we examine the case in which a recipient stops receiving transfers while others in the same community continue to receive them. This design isolates the effect of exit from the effect of prior receipt, and is well-suited to detecting backlash, which is more plausibly triggered by relative deprivation within a community than by the mere termination of payments. This situation is policy-relevant. Beneficiary rolls are periodically updated in most unconditional cash transfer programs, meaning that involuntary, mid-program exits of the kind we study are commonplace. Understanding how women’s empowerment and well-being evolve following such exits — and whether departing beneficiaries may require complementary support — is therefore of direct practical importance. Combining linked administrative and survey data, we are able to identify these effects empirically. We document precise null effects across a comprehensive set of measures of women’s economic, social and psychological well-being, providing reassurance that the benefits women experience during the transfer are not countervailed by backlash at program exit.

2 Context and Background

2.1 BISP Unconditional Cash Transfer Program

We study the Benazir Income Support Program (BISP), Pakistan’s unconditional cash transfer program that was rolled out nationwide in 2008 as a PKR 3,000 quarterly payment. The amount was periodically increased, given high rates of inflation in the country, and was PKR 8,750 (US\$ 156 PPP) quarterly in 2023.¹⁰ Transfers are made to low-income, ever-married women with a valid national identity card.

2.2 First Wave of Proxy Means Test

Program beneficiaries are identified using a Proxy Means Test (PMT). The initial round of the Proxy Means Test was based on a 2010-12 nationwide door-to-door census by the government, which gathered information on household assets and other characteristics for around 27 million households – 87% of all households in Pakistan (Haseeb and Vyborny, 2022; Guven *et al.*, 2024). A formula was developed with technical support from the World Bank to proxy for household consumption, which assigned households a score ranging from 0 to 100 based on 23 poverty indicators from three broad categories: household and individual characteristics, ownership of durable goods and housing characteristics, and ownership of productive assets, particularly

¹⁰USD values are calculated at purchasing power parity, using the World Bank PPP conversion factor for GDP for LCU/USD see [World Bank \(2026\)](#).

landholding, livestock, and farm equipment (Churchill *et al.*, 2024). The formula was not known to field-level BISP officials, and the cutoff was selected *after* the census data collection was complete, based on budgetary constraints.

2.3 Updating the Proxy Means Test

In 2016, another nationwide door-to-door census was launched to update the data to be used as input in the new PMT (Guven *et al.*, 2024). The census was concluded in 2020 and covered nearly 35 million households – 93% of all households in Pakistan. The proxy means test formula was updated, again with the objective of predicting consumption based on more recent household consumption survey data, and re-calculated with the updated census data. A new eligibility cutoff was established in November 2021. Thus, households fell into one of four groups (illustrated in Figure OA1): those who will continue to receive the transfers (*always eligible*), those who will newly start receiving transfers, households who will continue to not receive transfers, and households who will stop receiving transfers (*newly ineligible*). This study focuses on the always eligible and newly ineligible households. Importantly, the *newly ineligible* were not informed of their exit from the program; they learned about their new status only when they failed to receive their first post-reform payment and contacted BISP to inquire. Figure OA2 shows a timeline of these events.

In both the original and revised targeting exercises, the cutoff was established on the basis of budgetary constraints. Given the government’s limited budget, the eligibility threshold was determined post-data collection by finalizing the program’s budget, identifying a target number of recipients, and setting the threshold to cover that target population. This approach is considered to minimize the potential for manipulation around the cutoff, allowing for a clean regression discontinuity estimate (Camacho and Conover, 2011).

3 Data

To identify the effects of being discontinued from receiving cash transfers, we collect panel data on BISP recipients before and after the reform. We sample households from district Layyah, in southern Punjab, one of the poorest districts in the province with about 46% of the households categorised as ‘poor’ by the government of Pakistan (Government of Pakistan, 2016). The census activities for updating the Proxy Means Test for BISP were completed in Layyah district in 2020. Our sample is primarily composed of rural areas

that cover 11 Union Councils in Layyah.

To quantify the impact of discontinuation of the BISP cash transfer, we sample households from the new PMT census database after it had been updated. Our sample consisted of households that were (i) previously eligible for BISP given the old cutoff, and (ii) fell within a bandwidth of 3.5 points above and below the *new* cutoff. We screened out any recipients who reported not being BISP cash transfer beneficiaries in an initial screener question. This yields a sample of 2,449 recipients of whom 1,336 are always eligible and 1,113 are expected to become newly ineligible. We conduct a baseline survey with these recipients before any information about the discontinuation process had been released. In order to consistently measure intra-household bargaining and violence outcomes for women, we also screened out female recipients who were not partnered at that time.

The baseline survey was conducted in December 2021-January 2022, before the discontinuation of BISP payments in March 2022. We conduct a follow-up survey in person in February-March 2023, about a year after discontinuation. At this time, the discontinued households had not received two bi-annual cash transfer tranches.

Attrition in the follow-up survey was low, only 4.7%, and is balanced across the two groups of interest, the always eligible and the newly ineligible (Table OA1). As a result, we have a final analysis sample comprising a balanced panel of 2,333 households.

3.1 Sample Characteristics and Women's Outcomes

Table 1 presents descriptive statistics and baseline balance for our study sample of 2,449 households, of which 1,336 are classified as *always eligible* and 1,113 as *newly ineligible*. The two groups are largely similar in household and respondents' observable characteristics. The average age of the respondents is around 46 years. The rates of illiteracy are high in both groups, with 84-85% of the respondents being illiterate.

The average household has 6.5 members, reports a monthly expenditure of 16,000 PKR, and spends about 3,000 PKR on food (Panel A). Reported total expenditures are broadly in line with reported incomes¹¹, translating to roughly USD\$2 PPP per capita per day. Because these figures exclude the value of the consumption of own food produced, we do not benchmark them directly against the World Bank's extreme or moderate poverty lines. Nevertheless, they suggest that households in the sample are relatively poor.

¹¹Since the focus of the surveys was to measure women's empowerment, we did not thoroughly measure income. For rural households, where income streams are noisy, it would have required several questions to measure income by activity. Instead, we asked the respondent to report the typical monthly income of the household directly.

BISP transfers account for approximately 15% of household expenditures; because a large fraction of the consumption of rural households in our setting is from own production, transfers represent a smaller fraction of consumption. While we did not administer a comprehensive consumption module, consumption in the latest round of Household Integrated Economic Survey (2018-19) for BISP recipients in rural Punjab averages PKR 273,820 per year, suggesting that BISP payments would be 7.3% of consumption. Thus, BISP cash transfers represent a modest source of support. However, over half the women in the sample (54%) say that they decide *alone* how the funds from the BISP transfer are spent. This is striking evidence of the importance of the transfer in recipients' agency over spending, since women in this population often have neither income of their own that they can control, nor any say in the earnings of other household members. Specifically, 88% of our sample do not work for pay and thus have no earnings of their own.¹² Moreover, 47% of Pakistan DHS respondents indicate that the husband alone decides how his earnings are spent without input from their wives, meaning that a wife with no earnings of her own would have no say whatsoever in the expenditure of any part of household income (NIPS, 2019). To examine women's empowerment-related outcomes, we generate a series of Anderson indices (Anderson, 2008) and these are reported along with their components in Table 1 Panel B. We also show the impact on the key indices as RDD plots (Figures OA3 - OA4). Overall, the two groups are well balanced: the joint test of significance across the indices yields an F-statistic of $F_{10,2308} = 1.239, p = 0.261$. The only notable imbalance is in women's economic participation. However, our RDD-in-differences specification explicitly accounts for this imbalance, providing an advantage over a standard RDD approach.

We measure women's say in household decision-making as the extent of the respondent's involvement in decisions related to personal health, small household purchases, and small personal purchases. The responses are on a Likert scale from "Not at all" (1) to "High extent" (4). The decision-making score is nearly identical across both groups, on all these dimensions, averaging around 2.7: respondents have a small to medium extent of say in these decisions.

We measure women's mobility, as BISP beneficiaries have to go to the market for biometric verification to collect the BISP payments (Clark *et al.*, 2022), and evaluations of the program have indicated an increase in women's independent mobility (Cheema *et al.*, 2016; Ambler and de Brauw, 2024). About forty percent of the women report being able to go alone to the market, and about half can go with a friend. A similar

¹²A much larger fraction work unpaid, with nearly 4 hours spent on average on unpaid work on productive activities in the previous day.

proportion can go to a doctor without a family member. Reported mobility to go see family and friends is higher. Along all these dimensions of mobility, there are no significant differences at baseline between the two groups.

To capture potential backlash by partners who blame their wives for program exit or suspect them of hiding the funds, we capture measures of relationship quality, intimate partner violence and women's mental health outcomes. We ask women on a Likert scale of "Never" (1) to "Always" (4) whether the husband shows interest, respects them, and if they can talk to them about their problems. Along all these dimensions, women in both groups report that this happens often (values around 3), indicating a reasonable level of relationship quality with partners at baseline.

We measure the frequency of physical and sexual violence using the standard Demographic and Health Survey questions, and administer them using Audio Computer Assisted Self-Interviewing (ACASI) to protect the respondent's privacy.¹³ Physical and sexual violence is measured as a response to 14 questions about the frequency of each form of violence by the husband over the last 6 months on a scale of "Never" (1), "Once or Twice" (2), to "More Than Thrice" (3).¹⁴ We also ask about four types of injuries sustained as a result of something the partner did. We ask questions about emotional and controlling violence face-to-face.¹⁵ We use their responses to construct an indicator variable for the woman experiencing any type of violence in the last 6 months. 34% of the women across the two groups report experiencing some form of IPV. About 18% of women experience physical and 12% experience sexual violence. The rates are higher for emotional (22%) and controlling (19%) violence. Six percent report experiencing an injury because of an action by their partner.

Finally, we measure mental health and subjective well-being using several questions. Mental health is assessed using the Patient Health Questionnaire (the PHQ-9 score), which inquires about depression symptoms experienced in the last two weeks.¹⁶ Responses range from "Not at all" (0) to "Nearly every day" (3) and are summed to form a score. The average respondent has a score of 10.24, which corresponds

¹³ACASI is a self-interviewing method where women listen to audio recordings of questions in the local language on headphones, allowing them to privately respond to questions using a touch screen on a tablet. We developed an intuitive ACASI instrument with visual cues and piloted it extensively in our sample; we describe our data collection methodology in more detail in *Iqbal et al. (2025)*.

¹⁴These include getting pushed, slapped, arms twisted or hair pulled, punched, kicked or dragged, choked or burned, attacked, bruised, any dislocation or sprains, broken bones, forced to have sexual intercourse, forced to perform sexual acts, and facing violence in a state of intoxication from the husband.

¹⁵Emotional violence includes 3 questions about the frequency of humiliation, threats, or insults by the husband over the same period. Controlling violence includes questions about the frequency of accusations of infidelity, limiting contact with family, jealousy or anger over interactions with other men, or insistence on knowing her whereabouts.

¹⁶These include interest in life, feelings of hopelessness, sleep issues, tiredness, appetite changes, feelings of failure, concentration problems, slow movement or speech, and thoughts of self-harm.

to moderate symptoms of depression (Kroenke *et al.*, 2001). Correspondingly, the average score for life satisfaction (around 5.9) on a Likert scale of 1 (low) to 10 (high) and subjective happiness (around 3, corresponding to “Neither happy nor unhappy”) is quite low. Consistent with the low levels of reported expenditures and income, this indicates that the women had poor psychological well-being at baseline.

4 Empirical Strategy

Our main specification uses an RDD-in-differences approach to identify the effects of BISP program exit:

$$Y_{it} = \beta_0 + \beta_1 PMT2_i + \beta_2 NewlyIneligible_i + \beta_3 NewlyIneligible_i \times PMT2_i + \beta_4 Post_t + \beta_5 NewlyIneligible_i \times Post_t + \varepsilon_{it} \quad (1)$$

where Y_{it} is the outcome of interest for recipient i at time t . *NewlyIneligible* is an indicator variable for recipients who exited the program and stopped receiving transfers because they are above the new eligibility cutoff. *PMT2* is the running variable, a score based on the revised PMT in administrative data; we allow Y_{it} to vary with the running variable (β_1), to have a different level above and below the cutoff at baseline (β_2), a different relationship with the running variable above and below the cutoff (β_3), and to have a time trend independent of cash transfer status (β_4). Our coefficient of interest is β_5 : the effect of being above the new eligibility cutoff, after the cutoff was implemented, compared to those below the new cutoff who continued to receive the cash transfer.

Because the sample only includes households within +/- 3.5 points around the new eligibility cutoff on the PMT, it inherently includes a bandwidth around the RD cutoff. We also examine the robustness of results to a narrower bandwidth around the cutoff.

Importantly, Equation 1 estimates a Local Average Treatment Effect for those who ‘graduate’ from the program based on the updated Proxy Means Test. This corresponds to the impact of program graduation for recipients at relatively higher wealth levels among the poor. This treatment effect is of interest for policymakers considering program graduation models. However, it might be substantially different from the impact on the extremely poor if the program were fully discontinued; exit effects might be larger for women in much poorer households.

4.1 First Stage and Validity Checks

Regression discontinuity designs rely on discontinuous changes in program eligibility at a discrete point in the distribution of a smooth-running variable. We observe smoothness of the running variable around the new Proxy Means Test eligibility cutoff using the [Cattaneo *et al.* \(2018\)](#) test (Figure 1, Panel (a)) which finds no evidence of manipulation around the cutoff (p-value = 0.256).

We additionally augment our RD specification with a difference-in-differences approach using panel data on our sample. This allows us to account for any baseline differences in households across the two groups.

In principle, the receipt of the program should be driven by the Proxy Means Test score alone. This may not be the case for two reasons: first, some eligible women may not successfully sign up (non-compliance among eligible households). Second, women above the threshold can be exceptionally added to the beneficiary list due to exceptions to the eligibility criteria, in particular based on the presence of individuals with disabilities in the household ([Haseeb and Vyborny, 2022](#)), leading to non-compliance among households that would be ineligible based on the PMT. Thus, Equation 1 is an Intent-to-Treat specification. Nonetheless, we find that the official PMT cutoffs strongly predict inclusion and exclusion in the cash transfer. Figure 1 shows that the PMT2 RD cutoff strongly predicts household eligibility at follow-up. In the BISP administrative data, 7% of the respondents above the new PMT cutoff (newly ineligible) receive BISP after the policy change, while 88% of those below the cutoff (always eligible) do.

5 Results

5.1 Average Effects

We find that one year after exiting the BISP program, women faced no backlash or a decline in agency within the household compared to women who continued to receive the cash transfer (Tables 2-4). We also show this using RDD plots for the main outcomes in Figures OA5-OA6. For simplicity of presentation, we only show β_5 (Newly Ineligible $\times$ Post) in the tables. This captures the effect of being exited from the cash transfer program, versus continuing to receive cash transfers. It is estimated using equation 1 for the panel of 2,333 households that we survey both at baseline and at follow-up. Based on 95% confidence intervals, we can rule out impacts larger than 0.12 SD on the standardised indices used to capture women's empowerment (described in section 3.1). The null results also hold for a narrower bandwidth around the new eligibility

cutoff (Tables [OA2](#) - [OA4](#)).

5.1.1 Backlash

A key concern with exiting only a subset of women in the community from the cash transfer program is that recipients may be blamed for its termination or accused of concealing the funds, thereby leading to a risk of backlash or extractive violence. We find no evidence that program exit affected reported relationship quality with the spouse, including whether the husband shows interest in her or respects her (Table 2, Panel A).

If the husband thinks that the woman is still receiving the transfers and hiding the money, he may use violence to extract the resources. We measure different types of violence and find no impact on the incidence and frequency of any type of intimate partner violence or on injuries sustained as a result of violence (Table 2 Panel B).

We also examine whether potential backlash or the cessation of funds itself leads to a deterioration in women's mental health and subjective well-being. We find no effects: neither women's self-assessed well-being (happiness, life satisfaction, and level of stress) nor their mental health, measured using the validated 9-item Patient Health Questionnaire, shows significant changes (Table 2 Panel C).

Our findings are supported by women reporting not being blamed in our sample households for the program exit: we asked women "If BISP funds were late or less than expected, or no money came in the last 12 months, did your husband or in-laws blame you?" and nearly 94% of the women who have exited the program report never being blamed.

5.1.2 Erosion of Empowerment

Another concern with program exit is that the empowerment gains achieved through these transfers could be quickly reversed. For over a decade, women have received payments in their name, potentially strengthening their influence in household decision-making. However, we find no evidence of a change in a decision-making index that captures a woman's role in decisions regarding her own health care, small purchases for herself, and small purchases for others in the household a year after program exit (Table 3 Panel A).

Previous work on BISP recipients has documented positive impacts on women's mobility ([Ambler and de Brauw, 2024](#); [Cheema et al., 2016](#)). This is potentially driven by the requirement for in-person biometric verification at a payment agent to withdraw the transfer amount [Clark et al. \(2022\)](#). If this effect is simply a mechanical result of visiting the payment point to retrieve funds, program exit may erode mobility. Therefore,

we measure women's mobility as being able to go to the market, doctor, friends, and family, alone or in a group, without a family member accompanying them. We do not find any changes in independent mobility outside the household one year after program exit on any of these dimensions (Table 3 Panel A).

We also examine women's willingness to conceal money from their spouse as a proxy for resource control. Respondents were presented with a series of choices: keep PKR 400 for themselves or transfer a larger specified amount (ranging from PKR 400 to 1,600) to their spouse. From these choices, we construct a measure which captures the amount of money for the household that a respondent is willing to give up to retain the funds personally. Our analysis shows no significant change in this measure in response to program exit.

Next, we examine how women adjust financially to no longer receiving the cash transfers. Specifically, we assess whether they increase their labor supply in an effort to generate additional personal income to compensate for the lost income stream. As shown in Table 3, Panel B, we find no evidence of such adjustments: women are not more likely to enter the labor force, increase their hours of work outside the home, or change the time allocated to household chores. This is consistent with [Banerjee *et al.* \(2017\)](#), who re-analyze evidence from multiple countries and find no systematic effect of transfers on labor supply.

The study was not designed to capture the impact on household economic outcomes; we only observe crude expenditure measures, on which we find no significant impact of program exit. Given that the transfer is the only income that most recipients bring into the household personally, the loss of transfers might change the allocation of household expenditure toward them. We examine total expenditures on food and clothing – domains where women commonly make decisions – and find no effects on average. We also test whether effects might operate via reduced consumption of food among women or their children. Specifically, we analyze the frequency of consumption of higher-quality food items – meat, eggs, and fruit – over the past seven days, separately for the woman, her husband, and her children. As shown in Table 4, Panel A, we find no evidence of a change across any of these groups. Similarly, when examining clothing expenditures separately for women, husbands, and children, we again find no impact.

The loss of income personally controlled by the woman could result in a household pulling children out of school and into work. However, we find no significant effects on labor supply among those under 16 or on school enrollment across all age groups using data at the household member level (Table 4, Panel B). Consistent with this, education expenditures remain unchanged.

5.2 Heterogeneous Effects

Although the average impacts of the exit are small and insignificant on women’s empowerment, they may mask variation across groups. We therefore examine treatment effects in subgroups of women with certain characteristics or coming from certain types of households that are likely to respond more strongly to exiting the cash transfer program. The results are presented in Figures [OB1](#) - [OB5](#) in appendix B.

We first look at household characteristics, including above median values of household size and the ratio of cash transfer to expenditures, as higher values of these may indicate a greater dependence on the cash transfer. We also test whether the effects vary for households with above-median PMT1 scores – those relatively better off initially. Next, we examine subgroups of women with different levels of baseline empowerment – those with higher or lower say in decision-making, mobility, and economic participation – as well as those reporting higher or lower levels of spousal relationship quality. For all these subgroups, the estimated impacts of program exit remain small and insignificant.

6 Conclusion

We study the impact on women of exiting a long-running cash transfer program – Pakistan’s Benazir Income Support Program (BISP) – comparing women from households just below the revised eligibility criteria who continue to receive cash transfers to those who no longer receive them. One year after program exit, we find no significant impacts on a range of outcomes capturing various dimensions of women’s empowerment.

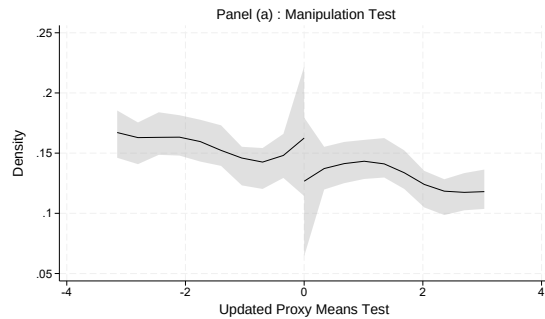
Since BISP cash transfers are directed to ever-married women, a potential concern is that program exit could trigger backlash against female recipients. However, we find no evidence of an increase in intimate partner violence, nor of any deterioration in women’s self-reported mental health or overall well-being. Taken together, our results suggest that the immediate negative consequences of program exit for women may be less severe than might be feared.

Because we estimate a Local Average Treatment Effect for those who ‘graduate’ from the program based on the updated Proxy Means Test, the effects we estimate are the impact of program graduation for recipients at relatively higher wealth levels among the poor. This treatment effect is of central interest for policymakers considering program exit and graduation. However, it is important to note that it might be substantially different from the impact on the extremely poor if the program were fully discontinued; exit effects might be larger for women in much poorer households.

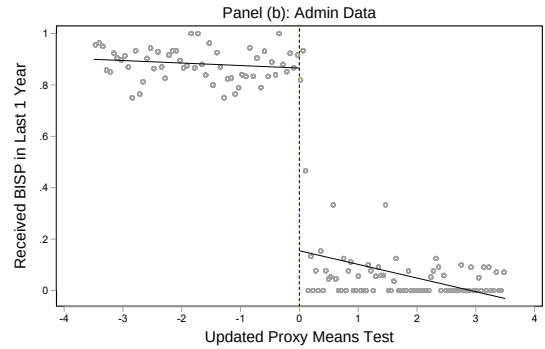
Table 1: Baseline descriptive Statistics and balance by BISP Eligibility Status

	<i>always eligible</i>	<i>newly ineligible</i>	mean difference	p-value
Panel A: Household & Respondent Characteristics				
Household Size	6.73	6.31	-0.42	0.00
Monthly Expenditure (PKR)	16271.39	16310.67	39.28	0.89
Total Food Expenditure (PKR/Week)	3088.05	3048.06	-39.99	0.71
Any Salaried Member	0.14	0.13	-0.01	0.46
Respondent's Age	45.45	46.31	0.86	0.00
Respondent Illiterate	0.85	0.84	-0.01	0.51
Panel B: Women's Outcomes				
Decision Making Index	0.00	0.01	0.01	0.75
Personal Healthcare (1-4)	2.77	2.78	0.02	0.59
Personal Small Purchases (1-4)	2.74	2.74	0.00	0.98
Household Small Purchases (1-4)	2.71	2.72	0.02	0.62
Mobility Index	-0.00	-0.05	-0.05	0.27
Market Alone (Indicator)	0.43	0.41	-0.03	0.18
Market Alone/Group (Indicator)	0.55	0.53	-0.02	0.35
Doctor Alone (Indicator)	0.49	0.47	-0.02	0.33
Doctor Alone/Group (Indicator)	0.58	0.56	-0.02	0.21
Friends Alone (Indicator)	0.83	0.82	-0.01	0.63
Friends Alone/Group (Indicator)	0.91	0.90	-0.01	0.42
Family Alone (Indicator)	0.67	0.67	-0.00	0.84
Family Alone/Group (Indicator)	0.75	0.73	-0.01	0.44
Economic Participation Index	-0.00	-0.09	-0.09	0.02
Employed (Indicator)	0.13	0.11	-0.02	0.10
Hours Worked (Per Week)	5.96	5.08	-0.88	0.15
Hours of Unpaid Work (Yesterday)	3.93	3.76	-0.17	0.08
Relationship Quality Index	0.00	-0.02	-0.02	0.65
Husband Shows Interest (1-4)	2.76	2.75	-0.01	0.75
Husband Respects(1-4)	2.98	2.97	-0.00	0.92
Can Talk About My Problems With Husband (1-4)	2.94	2.91	-0.03	0.45
IPV Index	0.00	0.05	0.05	0.26
Physical Frequency (1-3)	1.04	1.33	0.29	0.02
Sexual Frequency (1-3)	0.41	0.49	0.07	0.19
Emotional Frequency (1-3)	0.76	0.85	0.09	0.19
Controlling Frequency (1-3)	0.81	0.91	0.10	0.24
Injuries Frequency (1-3)	0.29	0.31	0.02	0.77
IPV Indicator	0.33	0.35	0.02	0.42
Physical Indicator	0.17	0.19	0.02	0.20
Sexual Indicator	0.11	0.13	0.01	0.40
Emotional Indicator	0.22	0.22	0.00	0.88
Controlling Indicator	0.19	0.20	0.01	0.70
Injuries Indicator	0.06	0.06	0.00	0.70
Patient Health Questionnaire (0-27)	10.24	10.24	0.00	1.00
Life Satisfaction (1-10)	5.95	5.82	-0.13	0.11
Subjective Happiness (1-5)	3.10	3.02	-0.08	0.03
Stress Index	-0.00	-0.03	-0.03	0.45
Able To Control Things (1-4)	2.79	2.78	-0.00	0.88
Confident In Handling Personal Problems (1-4)	2.22	2.20	-0.02	0.58
Things Going Your Way (1-4)	2.16	2.14	-0.03	0.40
Able To Cope With Difficulties (1-4)	2.74	2.75	0.01	0.86
Observations		1336	1113	

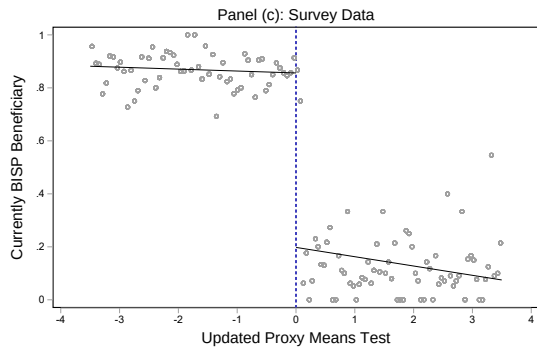
Note: The table reports the household and respondent characteristics at baseline separately for those who were discontinued (*newly ineligible*) and those who were not (*always eligible*). See appendix table OA1 for variable definitions. Mean difference is the difference between always eligible and newly ineligible and the reported p-values are for a test of the difference between the newly ineligible and always eligible.



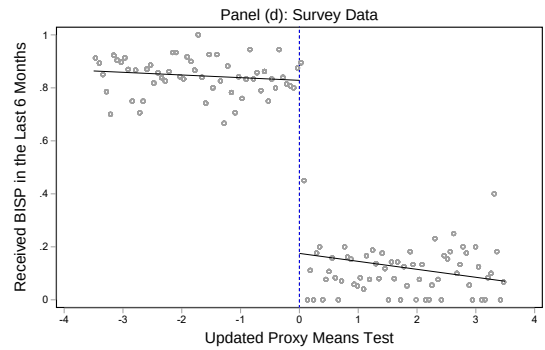
Sample below cutoff: 1278 ; Sample above cutoff: 1055 ;
p-value = 0.256.



Sample below cutoff: 1278; Sample above cutoff: 1055;
Bins below cutoff: 62; Bins above cutoff: 61.



Sample below cutoff: 1278; Sample above cutoff: 1055;
Bins below cutoff: 62; Bins above cutoff: 67.



Sample below cutoff: 1278; Sample above cutoff: 1055;
Bins below cutoff: 62; Bins above cutoff: 66.

Figure 1: RDD Validity Checks

Note: The figure in Panel (a) shows the Local Polynomial Density Estimator proposed in [Cattaneo et al. \(2020\)](#) for our analysis sample of 2,333 households. The local polynomial density estimate (solid black) and robust bias-corrected confidence intervals (shaded grey) are computed using the Stata package described in [Cattaneo et al. \(2018\)](#). The figures in Panel (b), Panel (c), and Panel (d) are generated using the `rdplot` command in Stata, as described in [Calonico et al. \(2014\)](#) to visualize the regression discontinuity design (RDD) with PMT as the running variable. The `rdplot` command uses evenly spaced bins to mimic the underlying variability of the data i.e. PMT is divided into bins of equal width, and each scatter-point represents the mean outcome variable in each of the bins, ensuring that the visual representation accurately reflects the distribution of the observations around the cutoff. The outcome variable in panel (b) “Received BISP in last 1 year” is an indicator variable for the woman having received BISP payments in the last 1 year, constructed using BISP admin data. The outcome variable in panel (c) “Current BISP beneficiary” is an indicator variable for the respondent reporting being a current BISP beneficiary at the follow-up survey round. The outcome variable in panel (d) “Received BISP In The Last 6 Months” is an indicator variable for the respondent reporting having received BISP in the last 6 months at the follow-up survey round. PMT2, the running variable is the Proxy Means Test assigned to each household. The PMT2 score has been standardised relative to the eligibility cutoff level, and the vertical dotted line at 0 indicates the eligibility cutoff. All households on the left side of this vertical line are considered BISP eligible, and all households on the right side are considered newly ineligible. The sample for all figures is 2,333 recipients that we successfully followed up with a year after the baseline. The solid black lines in both figures depict the first-order polynomial fit used to approximate the conditional expectations on both sides of the cutoff.

Table 2: Impact on Relationship Quality, IPV, and Well-Being

	(1) Newly Ineligible x Post	(2) SE	(3) Control Mean	(4) Obs	(5) HHs
Panel A: Relationship Quality					
Relationship Quality Index	-0.026	0.059	-0.000	4546	2332
Husband Shows Interest (1-4)	-0.031	0.052	2.698	4537	2330
Husband Respects (1-4)	-0.042	0.050	3.015	4537	2331
Can Talk About My Problems With Husband (1-4)	0.027	0.052	2.829	4528	2329
Panel B: Intimate Partner Violence (Last 6 Months)					
IPV Index	-0.064	0.058	0.000	4653	2328
Physical Frequency (1-3)	-0.188	0.160	0.555	4652	2328
Sexual Frequency (1-3)	-0.013	0.071	0.174	4641	2327
Emotional Frequency (1-3)	-0.036	0.110	0.687	4646	2328
Controlling Frequency (1-3)	-0.147	0.115	0.613	4646	2328
Injuries Frequency (1-3)	-0.032	0.069	0.136	4652	2328
IPV Indicator	-0.007	0.025	0.233	4653	2328
Physical Indicator	0.004	0.019	0.085	4652	2328
Sexual Indicator	0.007	0.016	0.045	4642	2328
Emotional Indicator	-0.002	0.022	0.144	4646	2328
Controlling Indicator	-0.021	0.022	0.149	4646	2328
Injuries Indicator	-0.010	0.012	0.031	4652	2328
Panel C: Mental Health And Subjective Well-Being					
Patient Health Questionnaire (0-27)	0.060	0.355	10.617	4666	2333
Life Satisfaction (1-10)	0.010	0.114	5.498	4661	2333
Subjective Happiness (1-5)	0.019	0.051	3.273	4658	2333
Stress Index	0.055	0.060	-0.000	4651	2329
Able To Control Things (1-4)	0.037	0.048	2.907	4647	2333
Confident In Handling Personal Problems (1-4)	0.022	0.045	1.936	4638	2333
Things Going Your Way (1-4)	0.010	0.045	1.839	4635	2333
Able to Cope with Difficulties (1-4)	-0.006	0.049	2.788	4644	2333

Note: The table reports results estimated using equation 1 for the panel of 2,333 households that we survey both at baseline and at the follow-up. We report the Newly Ineligible x Post (β_5) coefficient in column 1. Standard errors (SE) are reported in column 2. The mean for the dependent variable at follow-up for the control group (*always eligible*) is in column 3. See appendix OA1 for variable definitions. The number of observations varies because some respondents either declined to answer or reported not knowing.

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Impact on Agency and Economic Participation

	(1) Newly Ineligible x Post	(2) SE	(3) Control Mean	(4) Obs	(5) HHs
Panel A: Agency					
Decision Making Index	-0.007	0.059	-0.000	4666	2333
Personal Healthcare (1-4)	0.007	0.048	2.769	4664	2333
Personal Small Purchases (1-4)	-0.000	0.049	2.775	4635	2333
Household Small Purchases (1-4)	-0.032	0.049	2.822	4624	2333
Mobility Index	0.025	0.059	-0.000	4666	2333
Market Alone (Indicator)	0.009	0.027	0.293	4665	2333
Market Alone/Group (Indicator)	-0.010	0.028	0.409	4665	2333
Doctor Alone (Indicator)	0.017	0.029	0.358	4660	2333
Doctor Alone/Group (Indicator)	0.005	0.029	0.465	4660	2333
Friends Alone (Indicator)	0.020	0.023	0.829	4662	2333
Friends Alone/Group (Indicator)	0.013	0.017	0.912	4662	2333
Family Alone (Indicator)	0.003	0.028	0.588	4647	2333
Family Alone/Group (Indicator)	0.003	0.026	0.655	4647	2333
WTP To Hide Income From Spouse (PKR)	14.625	18.393	233.920	4582	2291
Panel B: Economic Participation and Time Use					
Economic Participation Index	0.054	0.054	-0.000	4666	2333
Employed (Indicator)	0.022	0.019	0.135	4666	2333
Hours Worked (Per Week)	0.584	0.781	10.078	4664	2332
Hours Of Unpaid Work (Yesterday)	0.045	0.128	2.659	4666	2333
Hours Chores (Yesterday)	-0.197	0.148	4.897	4666	2333

Note: The table reports results estimated using equation 1 for the panel of 2,333 households that we survey both at baseline and at the follow-up. We report the Newly Ineligible x Post (β_5) coefficient in column 1. Standard errors (SE) are reported in column 2. The mean for the dependent variable at follow-up for the control group (*always eligible*) is in column 3. See appendix OA1 for variable definitions. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Impact on Consumption and Child Outcomes

	(1) Newly Ineligible x Post	(2) SE	(3) Control Mean	(4) Obs	(5) HHs
Panel A: Intrahousehold Consumption Allocation					
Total Food Expenditure (PKR/Week)	100.003	106.330	4250.354	4646	2323
Husband High Quality Food Index	0.048	0.059	-0.000	4656	2328
Eats Meat (Times/Week)	0.249	0.315	0.452	4656	2328
Eats Eggs (Times/Week)	-0.206	0.283	0.542	4656	2328
Eats Fruits (Times/Week)	-0.288	0.303	0.625	4656	2328
Respondent High Quality Food Index	-0.067	0.056	0.000	4656	2328
Eats Meat (Times/Week)	-0.277	0.203	0.764	4656	2328
Eats Eggs (Times/Week)	-0.177	0.186	0.684	4656	2328
Eats Fruits (Times/Week)	0.274	0.209	0.704	4656	2328
Child High Quality Food Index	-0.011	0.073	0.000	2877	1830
Eats Meat (Times/Week)	-0.096	0.303	0.408	2875	1829
Eats Eggs (Times/Week)	0.511	0.356	0.279	2873	1829
Eats Fruits (Times/Week)	0.304	0.369	0.368	2874	1829
Total Clothing Expenditure (PKR/6 Months)	-174.664	233.458	3520.194	4666	2333
Husband	-44.387	38.986	488.772	4666	2333
Respondent	48.959	57.404	532.355	4666	2333
Child	-89.415	160.990	2513.459	4666	2333
Panel B: Children's Education And Child Labour					
Enrolled (Ages 3-24)	-0.005	0.010	0.427	16800	2267
Primary (Ages 3-10)	0.007	0.029	0.690	3059	901
Secondary (Ages 11-13)	0.013	0.026	0.408	2788	1102
Higher Secondary (Ages 14-24)	-0.002	0.007	0.069	10953	2108
Educational Expenditure (PKR/Year)	42.852	450.848	10411.003	15106	2267
Primary (Ages 3-10)	-382.892	769.916	9355.538	2373	901
Secondary (Ages 11-13)	1081.007	966.387	8637.677	1967	1102
Higher Secondary (Ages 14-24)	-19.045	328.283	3346.753	9116	2108
Child Labour (Ages 10-15)	-0.016	0.021	0.272	6174	1686

Note: The table reports results estimated using equation 1 for the panel of 2,333 households that we survey both at baseline and at the follow-up. We report the Newly Ineligible x Post (β_5) coefficient in column 1. Standard errors (SE) are reported in column 2. The mean for the dependent variable at follow-up for the control group (*always eligible*) is in column 3. See appendix OA1 for variable definitions. The number of observations varies because some respondents declined to answer, reported not knowing, or were not asked the question if they did not have children. In Panel B, the unit of analysis is the child, with sample sizes differing across age ranges. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

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Supplemental Appendix

A Additional Tables and Figures

Table OA1: Attrition

	(1) Surveyed at Follow-up
Above New Proxy Means Test Cutoff	-0.00870 (0.00869)
Control Mean	0.957
Households	2449

Note: The table reports results for differential attrition from a regression that includes all surveyed households at baseline within a +/- 3.5 bandwidth of the PMT2 score. The dependent variable is an indicator variable for the household being surveyed at the follow-up. The regression coefficient indicates the probability of a completed survey for households above the PMT2 Cutoff, i.e., the newly ineligible group at each stage. The mean outcome value for the always eligible group is also reported. Standard errors are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table OA2: Impact on Relationship Quality, IPV, and Well-Being For Narrower Bandwidth (+/- 3 points on PMT)

	(1) Newly Ineligible x Post	(2) SE	(3) Control Mean	(4) Obs	(5) HHs
Panel A: Relationship Quality					
Relationship Quality Index	-0.055	0.064	0.000	3884	1992
Husband Shows Interest (1-4)	-0.043	0.056	2.692	3876	1990
Husband Respects (1-4)	-0.054	0.054	3.012	3876	1991
Can Talk About My Problems With Husband (1-4)	-0.016	0.056	2.833	3868	1989
Panel B: Intimate Partner Violence (Last 6 Months)					
IPV Index	-0.054	0.060	0.000	3973	1988
Physical Frequency (1-3)	-0.206	0.170	0.565	3972	1988
Sexual Frequency (1-3)	-0.034	0.077	0.179	3961	1987
Emotional Frequency (1-3)	-0.047	0.118	0.698	3969	1988
Controlling Frequency (1-3)	-0.119	0.122	0.590	3969	1988
Injuries Frequency (1-3)	-0.060	0.074	0.146	3972	1988
IPV Indicator	-0.002	0.027	0.234	3973	1988
Physical Indicator	0.003	0.020	0.087	3972	1988
Sexual Indicator	-0.001	0.018	0.047	3962	1988
Emotional Indicator	0.005	0.024	0.141	3969	1988
Controlling Indicator	-0.019	0.024	0.148	3969	1988
Injuries Indicator	-0.012	0.013	0.032	3972	1988
Panel C: Mental Health And Subjective Well-Being					
Patient Health Questionnaire (0-27)	0.237	0.382	10.718	3986	1993
Life Satisfaction (1-10)	-0.006	0.123	5.514	3982	1993
Subjective Happiness (1-5)	0.090	0.056	3.253	3980	1993
Stress Index	0.057	0.064	0.000	3971	1989
Able To Control Things (1-4)	0.025	0.051	2.916	3968	1993
Confident In Handling Personal Problems (1-4)	0.028	0.048	1.928	3962	1993
Things Going Your Way (1-4)	0.034	0.049	1.836	3959	1993
Able to Cope with Difficulties (1-4)	-0.017	0.053	2.793	3966	1993

Note: The table reports results estimated using equation 1 for the panel of 1,993 households that we survey both at baseline and at the follow-up within +/- 3 bandwidth. We report the Newly Ineligible x Post (β_5) coefficient in column 1. Standard errors (SE) are reported in column 2. The mean for the dependent variable at follow-up for the control group (*always eligible*) is in column 3. See appendix OA1 for variable definitions. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table OA3: Impact on Agency and Economic Participation for Narrower Bandwidth (+/- 3 points on PMT)

	(1) Newly Ineligible x Post	(2) SE	(3) Control Mean	(4) Obs	(5) HHs
Panel A: Agency					
Decision Making Index	-0.014	0.064	0.000	3986	1993
Personal Healthcare (1-4)	0.003	0.052	2.773	3984	1993
Personal Small Purchases (1-4)	-0.011	0.053	2.777	3958	1993
Household Small Purchases (1-4)	-0.037	0.053	2.835	3949	1993
Mobility Index	0.029	0.063	0.000	3986	1993
Market Alone (Indicator)	0.017	0.030	0.291	3985	1993
Market Alone/Group (Indicator)	0.006	0.031	0.402	3985	1993
Doctor Alone (Indicator)	0.014	0.032	0.356	3980	1993
Doctor Alone/Group (Indicator)	0.011	0.031	0.458	3980	1993
Friends Alone (Indicator)	0.013	0.024	0.829	3982	1993
Friends Alone/Group (Indicator)	0.013	0.019	0.907	3982	1993
Family Alone (Indicator)	-0.009	0.030	0.587	3972	1993
Family Alone/Group (Indicator)	-0.006	0.028	0.652	3972	1993
WTP To Hide Income From Spouse (PKR)	33.972*	19.867	222.996	3910	1955
Panel B: Economic Participation and Time Use					
Economic Participation Index	0.036	0.058	0.000	3986	1993
Employed (Indicator)	0.021	0.020	0.130	3986	1993
Hours Worked (Per Week)	0.286	0.837	10.076	3984	1992
Hours Of Unpaid Work (Yesterday)	-0.012	0.140	2.692	3986	1993
Hours Chores (Yesterday)	-0.162	0.159	4.902	3986	1993

Note: The table reports results estimated using equation 1 for the panel of 1,993 households that we survey both at baseline and at the follow-up within +/- 3 bandwidth. We report the Newly Ineligible x Post (β_5) coefficient in column 1. Standard errors (SE) are reported in column 2. The mean for the dependent variable at follow-up for the control group (*always eligible*) is in column 3. See appendix OA1 for variable definitions. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table OA4: Impact on Consumption and Child Labour for Narrower Bandwidth (+/- 3 points on PMT)

	(1) Newly Ineligible x Post	(2) SE	(3) Control Mean	(4) Obs	(5) HHs
Panel A: Intrahousehold Consumption Allocation					
Total Food Expenditure (PKR/Week)	127.725	115.320	4238.988	3968	1984
Husband High Quality Food Index	0.066	0.064	-0.000	3976	1988
Eats Meat (Times/Week)	0.305	0.342	0.477	3976	1988
Eats Eggs (Times/Week)	-0.348	0.315	0.607	3976	1988
Eats Fruits (Times/Week)	-0.329	0.325	0.684	3976	1988
Respondent High Quality Food Index	-0.070	0.061	-0.000	3976	1988
Eats Meat (Times/Week)	-0.302	0.235	0.756	3976	1988
Eats Eggs (Times/Week)	-0.105	0.185	0.681	3976	1988
Eats Fruits (Times/Week)	0.320	0.240	0.681	3976	1988
Child High Quality Food Index	-0.025	0.079	0.000	2458	1564
Eats Meat (Times/Week)	-0.074	0.354	0.327	2456	1563
Eats Eggs (Times/Week)	0.601	0.413	0.195	2456	1564
Eats Fruits (Times/Week)	0.349	0.421	0.280	2455	1563
Total Clothing Expenditure (PKR/6 Months)	-175.967	253.740	3472.794	3986	1993
Husband	-49.235	42.171	483.830	3986	1993
Respondent	45.776	62.082	527.353	3986	1993
Child	-73.343	175.000	2471.948	3986	1993
Panel B: Children's Education And Child Labour					
Enrolled (Ages 3-24)	-0.005	0.011	0.421	14340	1940
Primary (Ages 3-10)	-0.007	0.031	0.695	2615	768
Secondary (Ages 11-13)	0.009	0.029	0.401	2354	937
Higher Secondary (Ages 14-24)	-0.001	0.008	0.066	9371	1804
Educational Expenditure (PKR/Year)	215.219	482.625	10313.891	12875	1940
Primary (Ages 3-10)	-755.923	812.675	9578.569	2020	768
Secondary (Ages 11-13)	598.787	1045.581	8479.531	1658	937
Higher Secondary (Ages 14-24)	128.016	380.863	3423.570	7804	1804
Child Labour (Ages 10-15)	-0.011	0.023	0.270	5230	1435

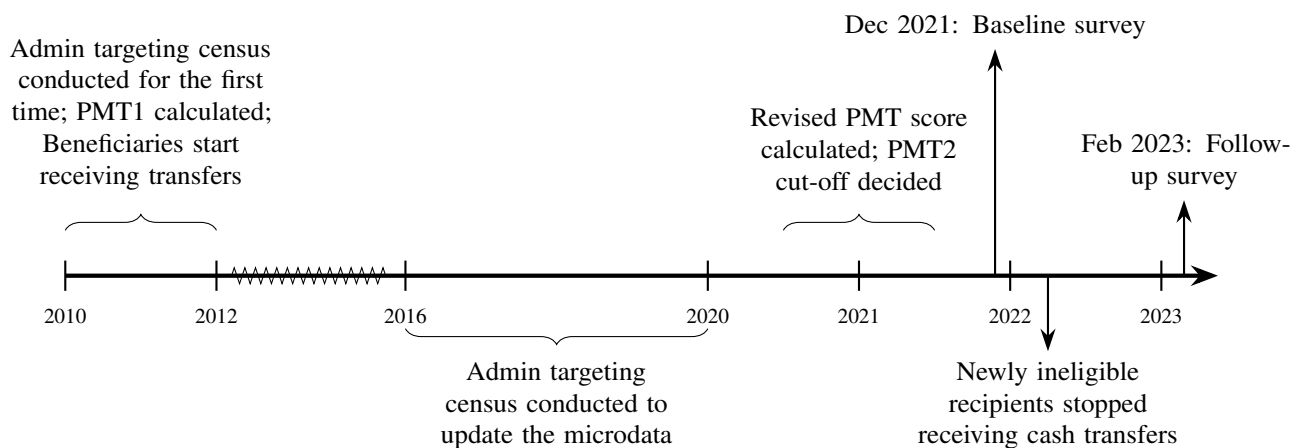
Note: The table reports results estimated using equation 1 for the panel of 1,993 households that we survey both at baseline and at the follow-up within +/- 3 bandwidth. We report the Newly Ineligible x Post (β_5) coefficient in column 1. Standard errors (SE) are reported in column 2. The mean for the dependent variable at follow-up for the control group (*always eligible*) is in column 3. See appendix OA1 for variable definitions. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Figure OA1: Scatter plot of households based on Proxy Means Test I and Proxy Means Test II scores



Notes: The two-way scatterplot shows proxy mean test scores (PMT) for households in Layyah calculated using 2010-11 (PMT1) and 2018-20 (PMT2) nationwide surveys. Each point on the scatterplot represents a household with corresponding PMT1 on the y-axis and PMT2 on the x-axis. The vertical line shows the PMT2 cutoff level and the horizontal line shows the PMT1 cutoff level. Consequently, four groups emerge. The two groups of interest for the paper are: households which are below the PMT1 and PMT2 cutoff i.e. *the always eligible* group, and households which are below PMT1 but above PMT2 i.e. *the newly ineligible* group

Figure OA2: Study Timeline



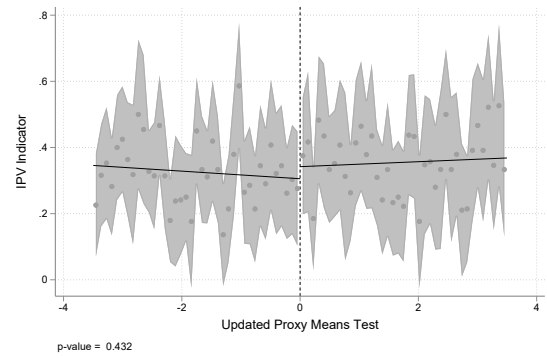
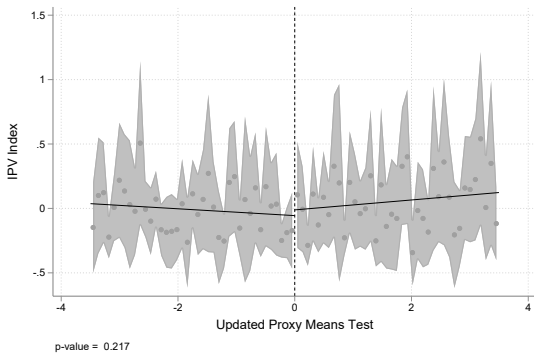
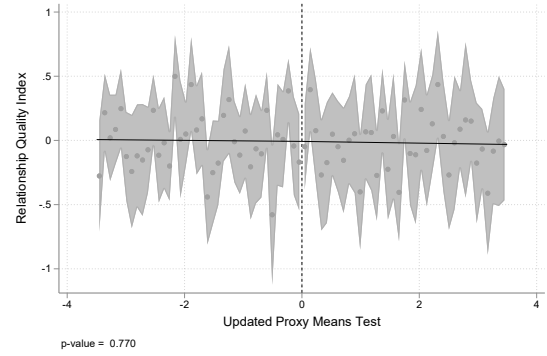
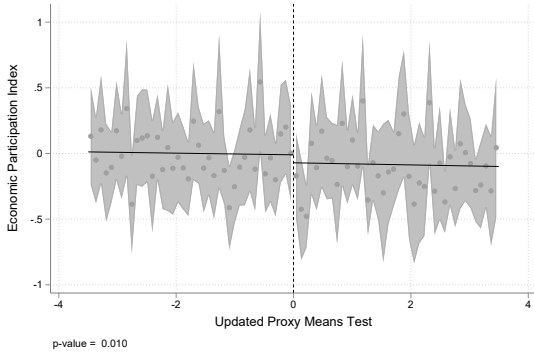
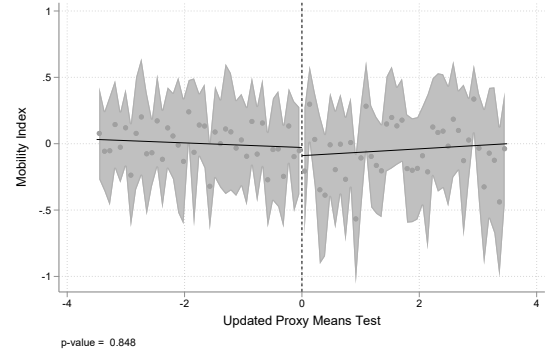
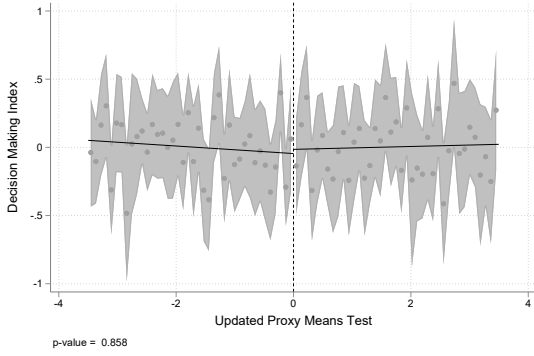


Figure OA3: RDD Validity Checks for Outcomes at Baseline - Part I

Note: The figures are generated using the `rdplot` command in Stata, as described in [Calonico *et al.* \(2014\)](#) to visualize the regression discontinuity with PMT as the running variable in the baseline outcomes. The sample for all figures consists of the 2,449 respondents to the baseline survey. The solid black lines in both figures depict the first-order polynomial fit used to approximate the conditional expectations on both sides of the cutoff.

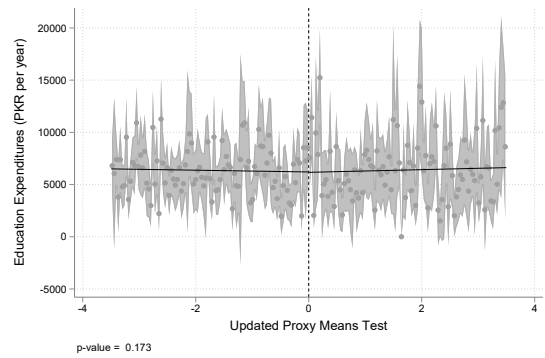
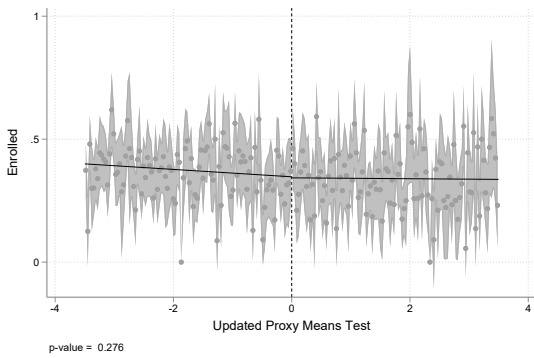
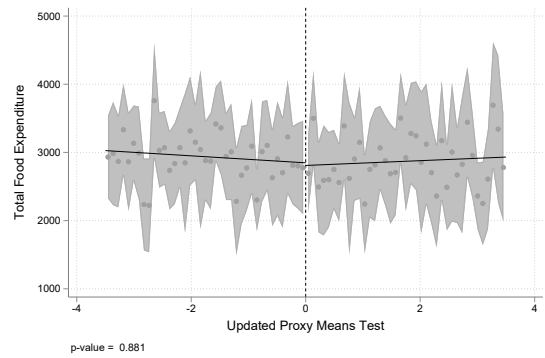
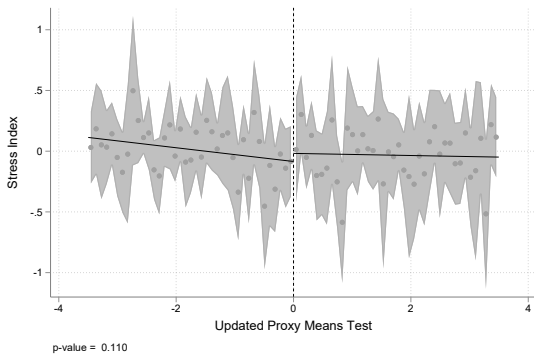


Figure OA4: RDD Validity Checks for Outcomes at Baseline - Part II

Note: The figures are generated using the `rdplot` command in Stata, as described in [Calonico *et al.* \(2014\)](#) to visualize the regression discontinuity with PMT as the running variable in the baseline outcomes. The sample for all figures consists of the 2,449 respondents to the baseline survey. The solid black lines in both figures depict the first-order polynomial fit used to approximate the conditional expectations on both sides of the cutoff.

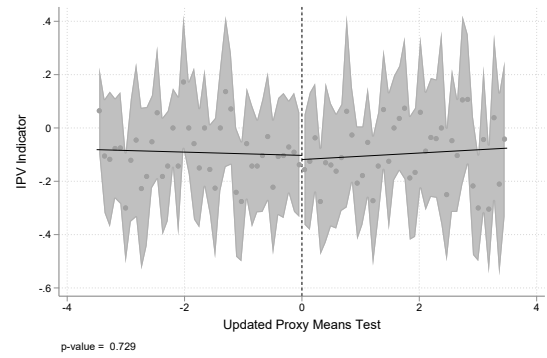
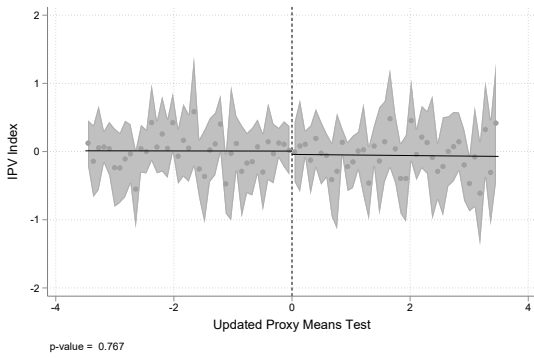
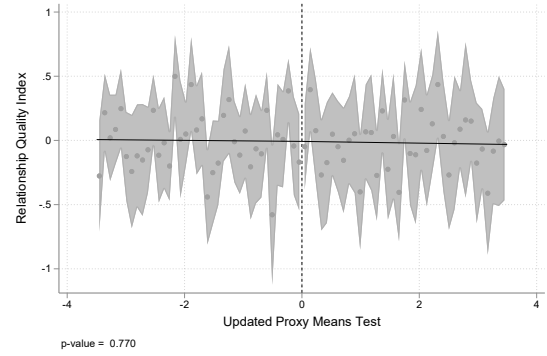
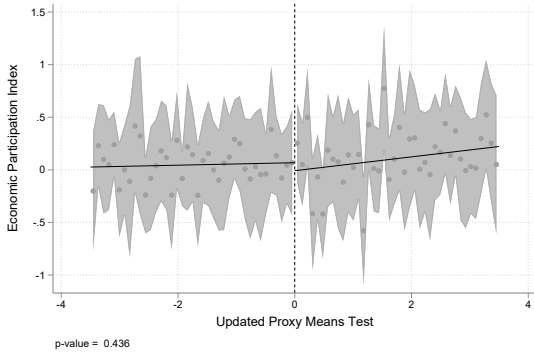
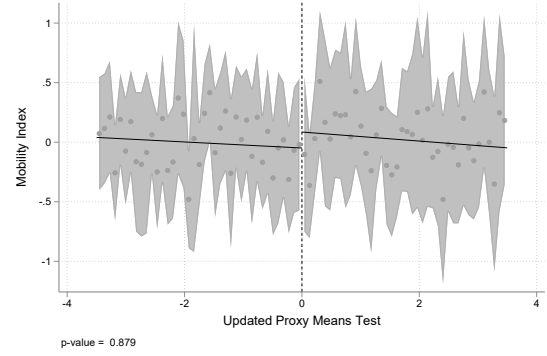
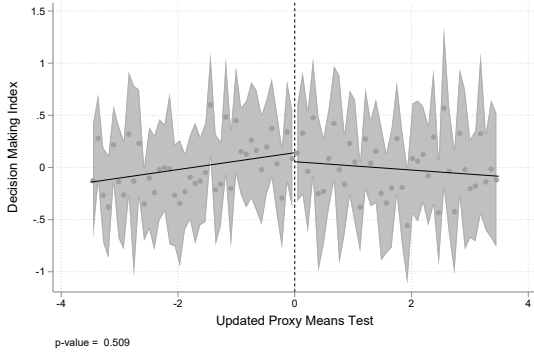


Figure OA5: RDD Plots for Outcomes at Endline - Part I

Note: The figures are generated using the `rdplot` command in Stata, as described in [Calonico *et al.* \(2014\)](#) to visualize the regression discontinuity with PMT as the running variable in the difference between the endline and baseline outcomes. The sample for all figures consists of the 2,333 respondents to the endline survey. The solid black lines in both figures depict the first-order polynomial fit used to approximate the conditional expectations on both sides of the cutoff.

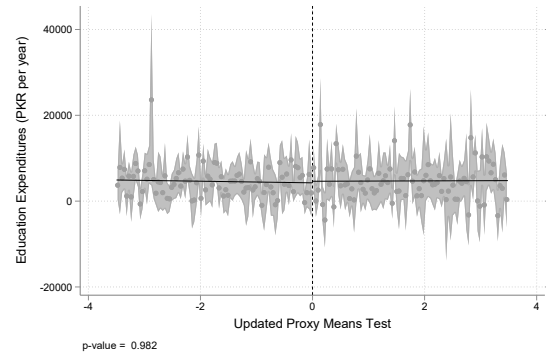
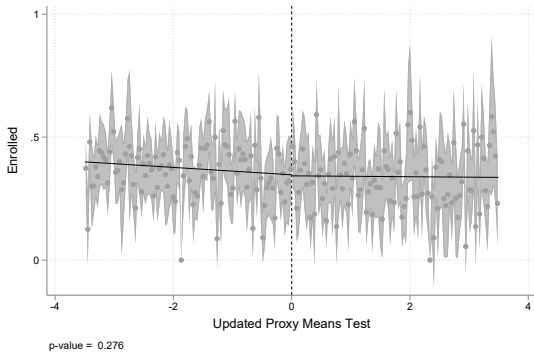
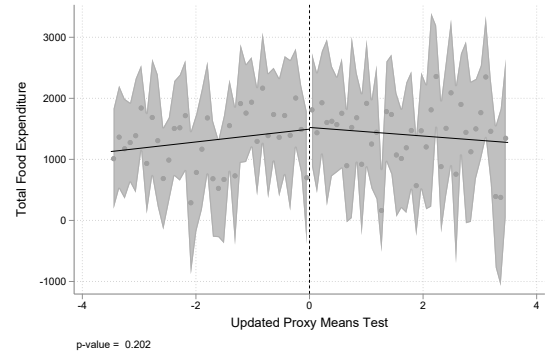
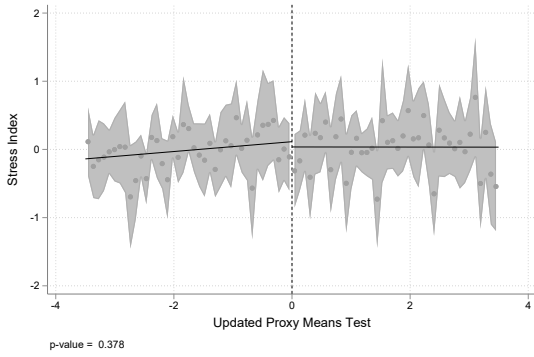


Figure OA6: RDD Plots for Outcomes at Endline - Part II

Note: The figures are generated using the `rdplot` command in Stata, as described in [Calonico *et al.* \(2014\)](#) to visualize the regression discontinuity with PMT as the running variable in the difference between the endline and baseline outcomes. The sample for all figures consists of the 2,333 respondents to the endline survey. The solid black lines in both figures depict the first-order polynomial fit used to approximate the conditional expectations on both sides of the cutoff.

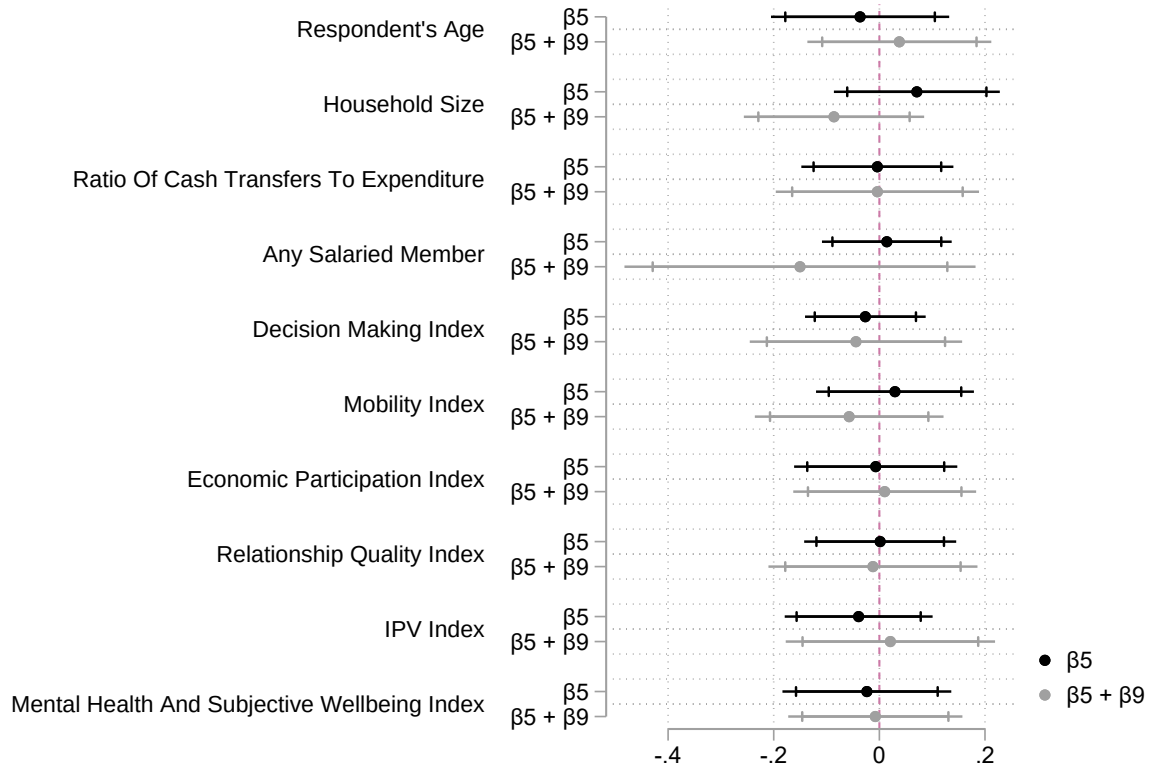
B Heterogeneous Treatment Effects

To estimate the heterogeneous impact of program exit, we augment equation 1 with H , the baseline dimension of interest:

$$Y_{it} = \beta_0 + \beta_1 PMT2_i + \beta_2 NewlyIneligible_i + \beta_3 NewlyIneligible_i \times PMT2_i + \beta_4 Post_t + \beta_5 NewlyIneligible_i \times Post_t + \beta_6 H + \beta_7 NewlyIneligible_i \times H + \beta_8 Post_t \times H + \beta_9 NewlyIneligible_i \times Post_t \times H + \varepsilon_{it} \quad (2)$$

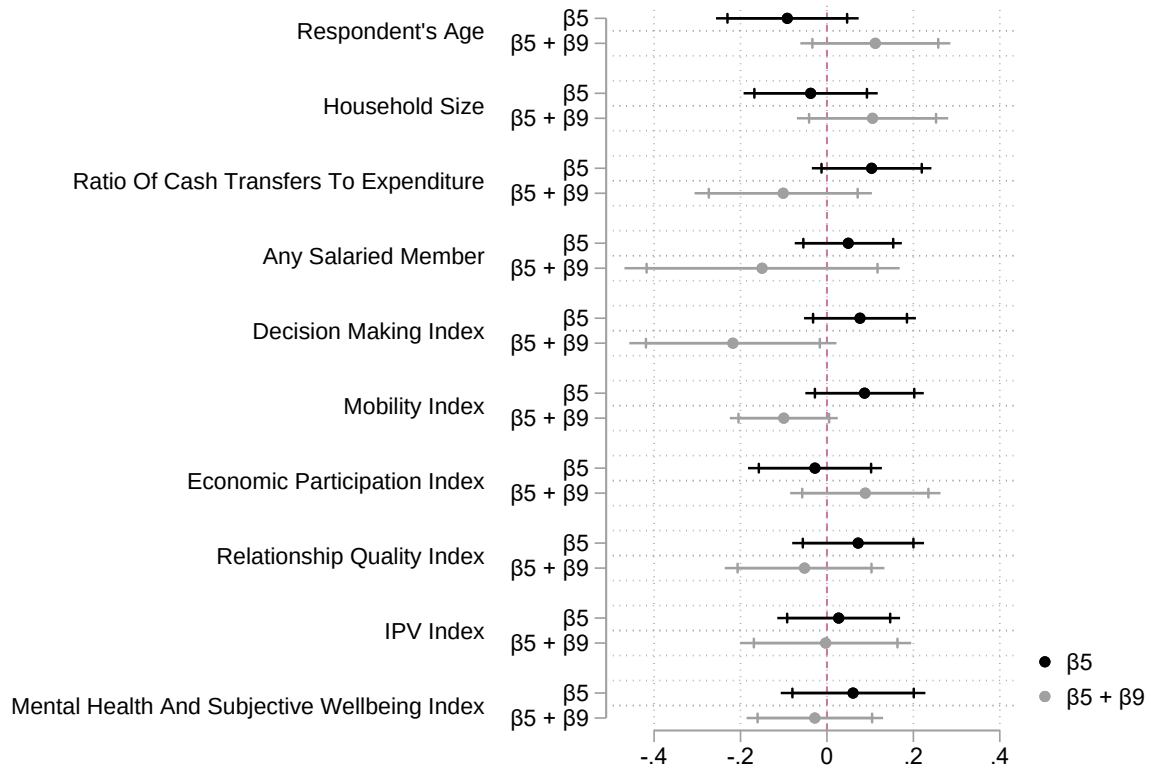
The coefficient of interest is β_9 which captures the differential impact of program exit for households on the dimension H at baseline. We convert all continuous measures into indicator variables equal to one for the above median value for the dimension at baseline. In Figures OB1 - OB5, for each dimension, we show both β_5 , the average effect on the newly ineligible and $\beta_5 + \beta_9$, which is the total effect on the newly ineligible with above median values at baseline.

Figure OB1: Heterogeneous Impact on the Decision Making Index



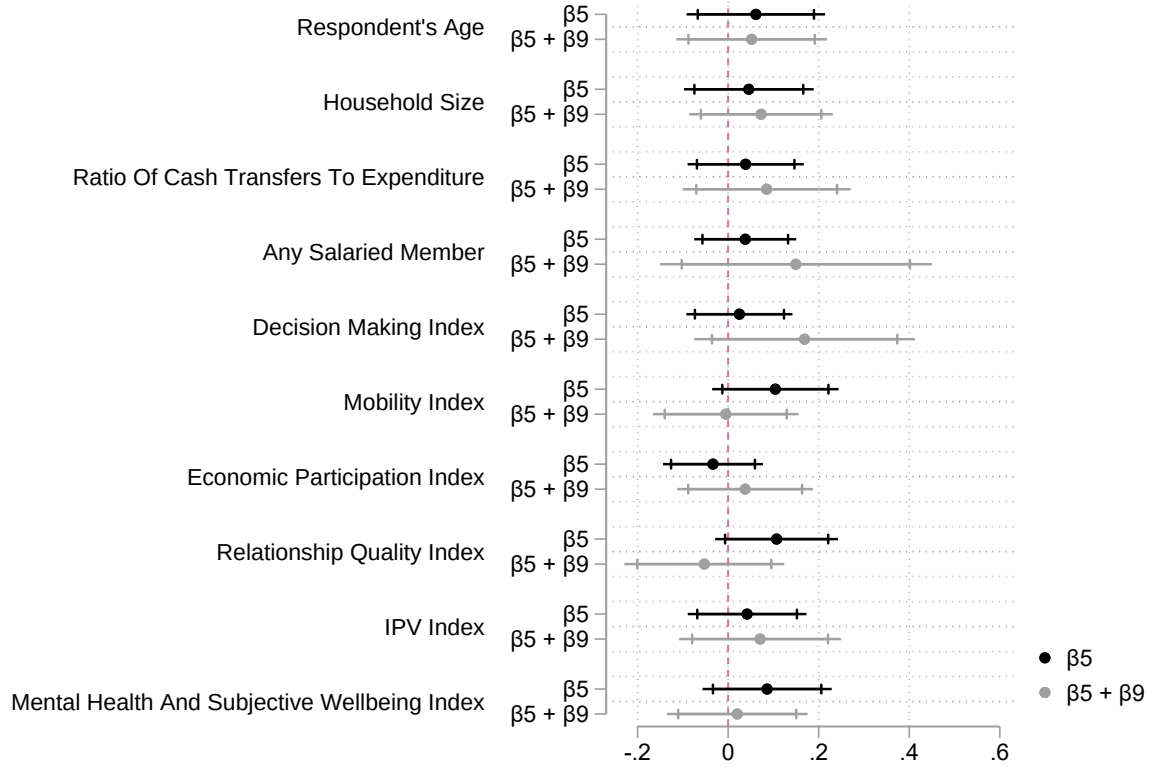
Notes: The dots are β_5 and $\beta_5 + \beta_9$ coefficients estimated using equation 2 for each heterogeneity dimension mentioned on the y-axis. The bars represent 90 and 95 percent confidence intervals. The sample used is a panel of 2,333 households that we survey both at baseline and at the follow-up. Each heterogeneity dimension is a binary indicator for above median value for the dimension at baseline. See appendix OA1 for variable definitions.

Figure OB2: Heterogeneous Impact on the Mobility Index



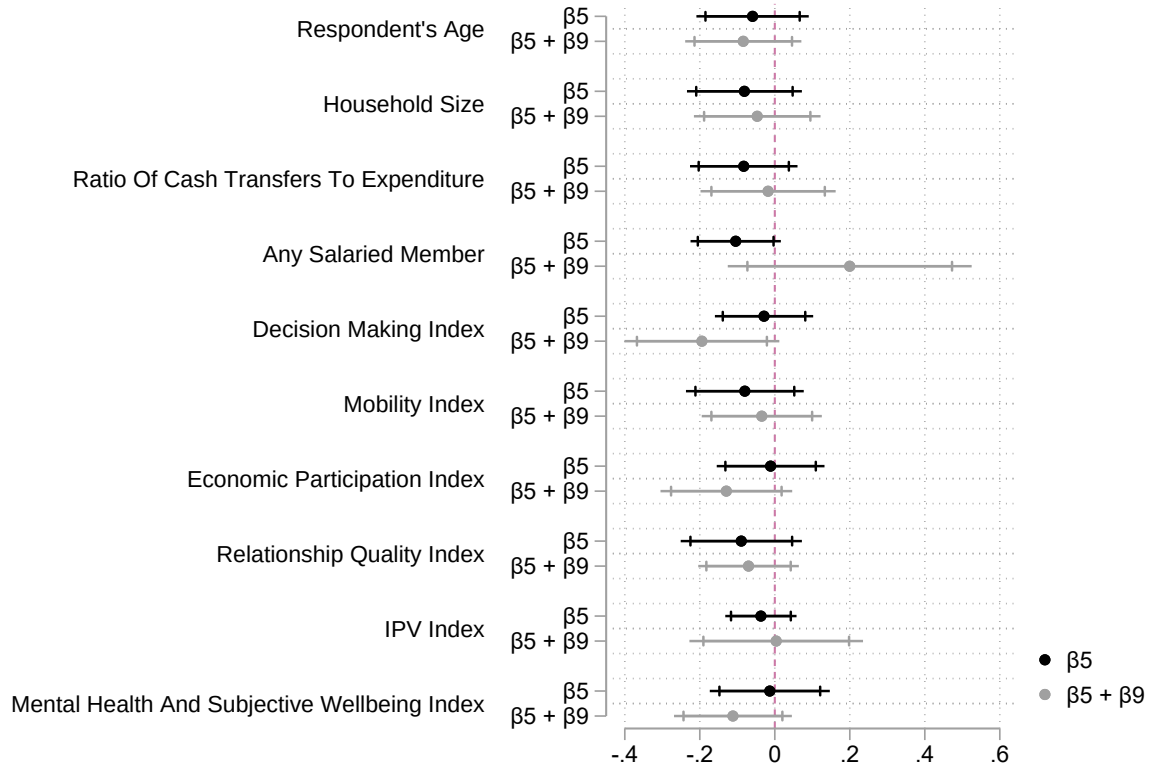
Notes: The dots are β_5 and $\beta_5 + \beta_9$ coefficients estimated using equation 2 for each heterogeneity dimension mentioned on the y-axis. The bars represent 90 and 95 percent confidence intervals. The sample used is a panel of 2,333 households that we survey both at baseline and at the follow-up. Each heterogeneity dimension is a binary indicator for above median value for the dimension at baseline. See appendix OA1 for variable definitions.

Figure OB3: Heterogeneous Impact on the Economic Participation Index



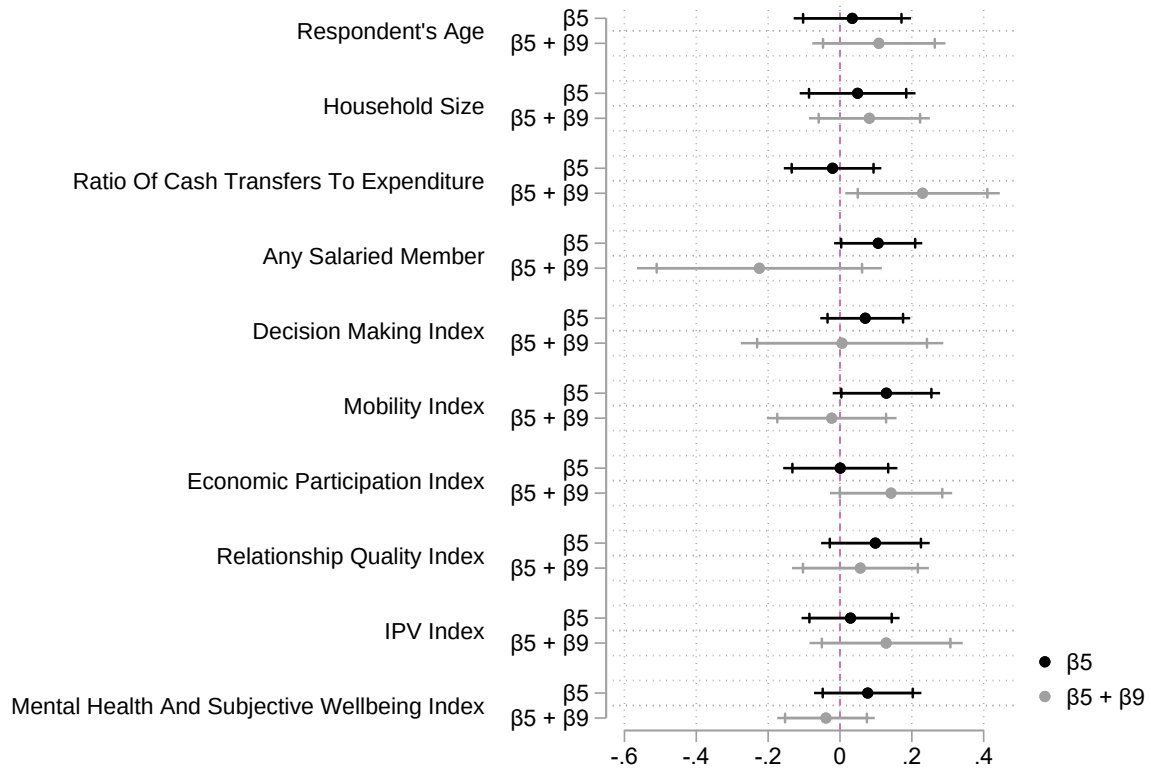
Notes: The dots are β_5 and $\beta_5 + \beta_9$ coefficients estimated using equation 2 for each heterogeneity dimension mentioned on the y-axis. The bars represent 90 and 95 percent confidence intervals. The sample used is a panel of 2,333 households that we survey both at baseline and at the follow-up. Each heterogeneity dimension is a binary indicator for above median value for the dimension at baseline. See appendix OA1 for variable definitions.

Figure OB4: Heterogeneous Impact on the Intimate Partner Violence Index



Notes: The dots are β_5 and $\beta_5 + \beta_9$ coefficients estimated using equation 2 for each heterogeneity dimension mentioned on the y-axis. The bars represent 90 and 95 percent confidence intervals. The sample used is a panel of 2,333 households that we survey both at baseline and at the follow-up. Each heterogeneity dimension is a binary indicator for above median value for the dimension at baseline. See appendix OA1 for variable definitions.

Figure OB5: Heterogeneous Impact on the Mental Health and Subjective Well-Being Index



Notes: The dots are β_5 and $\beta_5 + \beta_9$ coefficients estimated using equation 2 for each heterogeneity dimension mentioned on the y-axis. The bars represent 90 and 95 percent confidence intervals. The sample used is a panel of 2,333 households that we survey both at baseline and at the follow-up. Each heterogeneity dimension is a binary indicator for above median value for the dimension at baseline. See appendix OA1 for variable definitions.

B Variable Definitions

Table OA1: Variable Definitions

Variable	Definition
Panel A: Household & Respondent Characteristics	
Household Size	Total number of household members who eat together.
Monthly Expenditure (PKR)	Total household monthly expenditure.
Any Salaried Member	Indicator variable equal to 1 if any household member is salaried; 0 otherwise.
Respondent's Age	Age of the respondent in years.
Respondent Illiterate	Indicator variable equal to 1 if the respondent cannot read or write; 0 otherwise.
Panel B: Outcome Variables	
<i>Note: All indices are constructed using the <code>swindex</code> command in Stata suggested in (Schwab et al., 2021), following (Anderson, 2008). The indices are normalized with reference to the control group, i.e always eligible group.</i>	
Relationship Quality Index	Anderson index capturing the respondent's perception of relationship quality with the following questions. All responses coded as "Never" (1), "Sometimes" (2), "Often" (3), "Always" (4). Based on the question: "My husband shows interest in things that are important to me." Higher value means more frequent interest shown.
a. Husband Shows Interest (1–4).	
b. Husband Respects (1–4).	Based on the question: "My husband respects me." Higher value means more frequent respect shown.
c. Can Talk About Problems (1–4).	Based on the question: "I can talk about my problems with my husband." A higher value means a greater ability to discuss problems openly.
IPV Index	Anderson index constructed from 14 questions on physical, sexual, emotional, controlling, and injury-related violence experienced in the last 6 months. All responses are recoded as "Never" (1), "Once or Twice" (1.5), "More Than Thrice" (3).
a. Physical Frequency (1-3)	Frequency of physical violence faced by the respondent in the last 6 months, including getting pushed, slapped, arms twisted or hair pulled, punched, kicked/dragged, choked/burned, or attacked.
b. Sexual Frequency (1-3)	Frequency of sexual violence in the last 6 months, including whether the husband forced the respondent to have sexual intercourse, forced her to perform sexual acts, or threatened her with violence to perform sexual acts that she did not want to.
c. Emotional Frequency (1-3)	Frequency of emotional abuse faced by the respondent, including getting humiliated in front of others, being threatened to hurt or harm someone close to her, or getting insulted by the husband in the last 6 months.
d. Controlling Frequency (1-3)	Frequency of instances when the husband accused the respondent of being unfaithful, tried to limit her contact with her family, was jealous or angry over interaction with other men, or insisted on knowing about her whereabouts in the last 6 months.

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Table OA1 – continued from previous page

Variable	Definition
e. Injuries Frequency (1-3)	Frequency of instances where the respondent ever got any injuries like cuts, bruises, aches, burns, dislocations, deep wounds, broken bones, intoxicated, violent behavior from her husband in the last 6 months.
IPV Indicator	Binary variable equal to 1 if the respondent reports any instance of IPV in the last 6 months.
a. Physical Indicator	Binary variable equal to 1 if the respondent reports any instance of physical violence in the last 6 months.
b. Sexual Indicator	Binary variable equal to 1 if the respondent reports any instance of sexual violence in the last 6 months.
c. Emotional Indicator	Binary variable equal to 1 if the respondent reports any instance of emotional abuse in the last 6 months.
d. Controlling Indicator	Binary variable equal to 1 if the respondent reports any instance of controlling behavior in the last 6 months.
e. Injuries Indicator	Binary variable equal to 1 if the respondent reports any injury resulting from IPV in the last 6 months.
Patient Health Questionnaire (0-27)	Sum of responses to nine depression-related questions over the last 2 weeks. Questions include having little interest in things, feeling hopeless, trouble falling asleep, having little energy, poor appetite or excessive eating, feeling bad about self, having trouble concentrating on things, moving or speaking slowly, and thoughts of self-harm. Responses coded “Not at all” (0) to “Nearly every day” (3). A higher value corresponds to worsening of mental health
Life Satisfaction (1-10)	Self-reported satisfaction with life, on a scale from “Completely dissatisfied” (1) to “Completely satisfied” (10).
Subjective Happiness (1-5)	Self-reported happiness in the last month, coded as “Unhappy” (1) to “Very happy” (5).
Stress Index	Anderson index constructed from the following four questions, capturing perceived stress. All responses coded as “Never” (1) to “Very Often” (4). A higher value corresponds to better outcome.
a. Able to Control Things (1–4)	Self-reported frequency of feeling able to control important things in life.
b. Confident in Handling Personal Problems (1–4)	Self-reported confidence in ability to handle your personal problems.
c. Things Going Your Way (1–4)	Self-reported perception that things are going their way.
d. Able to Cope with Difficulties (1–4)	Self-reported frequency of feeling that difficulties were piling up beyond the ability to cope.
Decision-making Index	Anderson index capturing the respondent’s self-reported say in household decision-making across the following domains. All responses are on a scale of “Not at all” (1) to “High extent” (4).
a. Personal Healthcare (1–4)	Respondent’s reported say in decisions regarding her own health.
b. Personal Small Purchases (1–4)	Respondent’s reported say in decisions regarding small personal purchases, such as clothing for herself.
c. Household Small Purchases (1–4)	Respondent’s reported say in decisions regarding small household purchases, such as kitchen utensils for other members of the household.
Mobility Index	Anderson index constructed using the six components as follows. All responses take value 1 if the respondent says she is allowed to go; 0 otherwise.
a. Market Alone (Indicator)	Binary indicator for the respondent allowed to go to the local market to buy things alone.
b. Market Alone/Group (Indicator)	Binary indicator for the respondent allowed to go to the local market to buy things alone or in a group.
c. Doctor Alone (Indicator)	Binary indicator for the respondent allowed to go to the local health facility or doctor alone.
d. Doctor Alone/Group (Indicator)	Binary indicator for the respondent allowed to go to the local health facility or doctor alone or in a group.
e. Friends Alone (Indicator)	Binary indicator for the respondent allowed to go to homes of friends in the neighbourhood alone.

Continued on next page

Table OA1 – continued from previous page

Variable	Definition
f. Friends Alone/Group (Indicator)	Binary indicator for the respondent allowed to go to homes of friends in the neighbourhood alone or in a group.
g. Family Alone (Indicator)	Binary indicator for the respondent allowed to visit family alone.
h. Family Alone/Group (Indicator)	Binary indicator for the respondent allowed to visit family alone or in a group.
WTP to Hide Income from Spouse (PKR)	The monetary gap at which a respondent is willing to forgo money for herself to benefit her spouse. A higher value indicates a propensity to keep money for herself.
Economic Participation Index	Anderson index containing information on the following components:
a. Employed (Indicator)	Binary variable equal to 1 if the respondent reported working in the last month.
b. Hours Worked (Per Week)	Number of hours the respondent spends in a week on any kind of work with or without pay.
c. Hours of Unpaid Work (Yesterday)	Number of hours spent by the respondent in the last 24 hours on income-generating activities without pay.
Hours Chores (Yesterday)	Number of hours the respondent spent in the last 24 hours on household chores.
Total Food Expenditure (PKR/Week)	Value of food items consumed by the household in the last 7 days (95 percent winsorized).
Husband High Quality Food Index	Anderson index constructed from the following components for the husband. All responses coded as “Never” (0), “Once or Twice” (1.5), and “More Than Thrice” (3).
a. (Husband) Eats Meat (Times/Week)	The frequency of meat consumed by the husband over the last 7 days.
b. (Husband) Eats Eggs (Times/Week)	The frequency of eggs consumed by the husband over the last 7 days.
c. (Husband) Eats Fruits (Times/Week)	The frequency of fruits consumed by the husband over the last 7 days.
Respondent High Quality Food Index	Anderson index constructed from the following components for the respondent. All responses coded as “Never” (0), “Once or Twice” (1.5), and “More Than Thrice” (3).
a. (Respondent) Eats Meat (Times/Week)	The frequency of meat consumed by the respondent over the last 7 days.
b. (Respondent) Eats Eggs (Times/Week)	The frequency of eggs consumed by the respondent over the last 7 days.
c. (Respondent) Eats Fruits (Times/Week)	The frequency of fruits consumed by the respondent over the last 7 days.
Child High Quality Food Index	Anderson index constructed from the following components for the children. All responses coded as “Never” (0), “Once or Twice” (1.5), and “More Than Thrice” (3).
a. (Child) Eats Meat (Times/Week)	The frequency of meat consumed by the respondent over the last 7 days.
b. (Child) Eats Eggs (Times/Week)	The frequency of eggs consumed by the respondent over the last 7 days.
c. (Child) Eats Fruits (Times/Week)	The frequency of fruits consumed by the respondent over the last 7 days.
Total Clothing Expenditure (PKR/6 Months)	Household spending on clothing for the respondent, her husband, and children in the last 6 months (95 percent winsorized).
a. Husband (Clothing)	Total household spending on the husband’s clothing in the last 6 months (95 percent winsorized).
a. Respondent (Clothing)	Total household spending on the respondent’s clothing in the last 6 months (95 percent winsorized).
a. Child (Clothing)	Total household spending on the child’s clothing in the last 6 months (95 percent winsorized).
Enrolled (Ages 3-24)	Binary variable for all individuals in a household aged 3–24 years enrolled in private or public education institutions.
a. Primary (Ages 3-10)	Binary variable for individuals aged 3–10 years enrolled in grades 1–5.
b. Secondary (Ages 11-13)	Binary variable for individuals aged 11–13 years enrolled in grades 6–10.

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Variable	Definition
c. Higher Secondary (Ages 14-24) Educational Expenditure (PKR/Year)	Binary variable for individuals aged 14–24 years enrolled in grades 11–12. Member-level yearly educational expenditure for all individuals aged 3–24 years enrolled in grades 1–12. Constructed by summing the following outcomes for all members of the household:
a. Primary (Ages 3-10)	Member-level yearly educational expenditure for individuals aged 3–10 years enrolled in grades 1–5.
b. Secondary (Ages 11-13)	Member-level yearly educational expenditure for individuals aged 11–13 years enrolled in grades 6–10.
c. Higher Secondary (Ages 14-24)	Member-level yearly educational expenditure for individuals aged 14–24 years enrolled in grades 11–12.
Child Labor (Ages 10-15)	Binary variable for all household members aged 10–15 years; equal to 1 if employed or seeking employment and 0 otherwise.

Panel C: Heterogeneous Treatment Effect (HTE) Variables

Note: All variables in this panel are constructed as binary indicators equal to 1 if above the sample median, and 0 otherwise.

Respondent's Age	Indicator for whether the respondent's age is above the median.
Household Size	Indicator for whether household size is above the median.
Ratio of Cash Transfers to Expenditure	Indicator for whether the ratio of BISP cash transfers to household consumption (excluding value of own production consumed) is above the median.
PMT1 Score	Indicator equal to 1 if the household's normalised Poverty Means Test (PMT) score from the 2010–2011 NSER round is above the median.
Any Salaried Member	Indicator for whether the household has at least one salaried member.
Decision Making Index	Indicator for whether the Decision Making Index is above the median.
Mobility Index	Indicator for whether the Mobility Index is above the median.
Economic Participation Index	Indicator for whether the Economic Participation Index is above the median.
Relationship Quality Index	Indicator for whether the Relationship Quality Index is above the median.
IPV Index	Indicator for whether the IPV Index is above the median.
Mental Health and Subjective Well-Being Index	Indicator for whether the Mental Health and Subjective Well-Being Index is above the median.